Software Requirements Specification

Project: Smart Krishi Platform

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1. Introduction

This Software Requirements Specification (SRS) outlines the architectural and functional requirements for Smart Krishi, an integrated digital agriculture ecosystem tailored to bridge the gap between scientific agricultural research and ground-level farming operations. Traditional portals lack personalized, data-driven decision support. Smart Krishi resolves this by providing an environment for farmers to access rule-based agronomic recommendations, live meteorological tracking, and scientific fertilizer dose calculations.

The platform converges multi-tier caching with live telemetry, dynamically aggregating verified commodity market prices and meteorological data with external APIs (like WeatherAPI.com and AGMARKNET) to generate a comprehensive, actionable agricultural decision support system.

2. System Overview & User Roles

Smart Krishi operates as a distributed multi-tier architecture orchestrated by a Java 17 and Spring Boot 3.3.4 backend, supported by Caffeine Cache. The frontend is delivered via HTML5, ES6+ JavaScript, and Bootstrap 5.3.

2.1 User Classes

- **Farmer (ROLE_FARMER):** The primary authenticated user. Capabilities include managing multi-plot farm records, receiving personalized crop, fertilizer, and irrigation schedules, and calculating farm budgets.
- **Agricultural Administrator (ROLE_ADMIN):** Administrative users with elevated privileges to monitor platform health, curate government scheme advisories, and manage APMC market rates.
- **Guest Reader (ROLE_GUEST):** Unauthenticated users who can access public crop recommendations, live weather telemetry, nearby mandi discovery, and price trend forecasting.

> **Architectural Note:** Role-Based Access Control (RBAC) is strictly enforced at the API gateway layer utilizing Spring Security 6 and stateless JSON Web Tokens (JWT).

3. Functional Requirements (Modules 1 - 9)

Module 1: Farmer Management & Land Profiling

Provides farmer identity management, JWT token issuance, and multi-plot farm record maintenance.

- **Authentication:** Validates credentials and issues secure, stateless JWTs with BCrypt password

Module 2: Precision Crop Recommendation Engine

Prescribes agronomically suitable crops based on chemical soil composition and environmental conditions.

- **Multi-Parameter Evaluation:** Analyzes 8 parameters (Nitrogen, Phosphorus, Potassium, soil pH, temperature, humidity, rainfall, and soil classification) against ICAR benchmarks.
- **Agronomic Output:** Returns primary and secondary crop choices with suitable seasons and initial fertilizer suggestions.

Module 3: Live Meteorological Telemetry & Smart Irrigation

Provides real-time location-based atmospheric data and water management guidance without simulated values.

- **Live Telemetry:** Interfaces directly with WeatherAPI.com to ingest real-time atmospheric readings and 5-day forecasts based on GPS coordinates.
- **Smart Irrigation Advisory:** Computes an evapotranspiration and soil moisture deficit index to recommend "PAUSE", "HEAVY", or "MODERATE" irrigation.

Module 4: Smart Fertilizer Recommendation Advisor

Calculates exact chemical fertilizer quantities aligned with ICAR standard practices and advises soil pH correction.

- **Nutrient Conversion:** Translates elemental N-P-K nutrient deficits into commercial Indian fertilizer bags (Urea, DAP, MOP, SSP, Zinc Sulphate).
- **pH Correction:** Advises agricultural lime for acidic soils (pH < 6.0) and gypsum for alkaline soils (pH > 8.0).

Module 5: Crop Profit & ROI Calculator

Provides harvest financial projections, cost structure breakdowns, and investment return metrics.

- **Financial Modeling:** Accepts costs for seeds, fertilizers, pesticides, labor, irrigation, and machinery to compute total production cost.
- **Profit Metrics:** Calculates net profit, ROI percentage, break-even selling price per quintal, and classifies venture health (e.g., EXCELLENT, LOW MARGIN).

Module 6: Crop Growth Calendar & Phenological Advisory

Generates a structured timeline of crop development stages with critical agricultural action items tailored to the real-world sowing date.

- **Phenological Stages:** Maps 6 discrete growth stages including Sowing, Germination, Vegetative/Tillering, Flowering, Grain Filling, and Harvesting.
- **Actionable Milestones:** Provides precise irrigation schedules, top-dressing milestones, and protective field precautions for each specific stage.

Module 7: Crop Price Trend Prediction Engine

Calculates short-to-medium term commodity price projections using historical APMC time-series data.

- **Statistical Modeling:** Utilizes a rolling simple moving average and linear momentum formulas to project price trends (RISING, FALLING, or STABLE).
- **Data Integrity:** Enforces a strict minimum of 3 historical data points to generate a forecast, otherwise returning a transparent INSUFFICIENT_DATA status.

Module 8: Nearby APMC Mandi Market Prices

Finds the closest operational APMC market yards and transparent market prices using live government data streams.

- **Geodesic Locator:** Implements the Haversine spherical trigonometric formula to filter nearest mandis within 10, 25, 50, or 100 km radius.
- **Data Aggregation:** Fetches live commodity mandi rates published by the Ministry of Agriculture (AGMARKNET data.gov.in API) with fallback layers.

Module 9: System Administration & Governance

Empowers platform administrators to oversee operations, publish market rates, and manage government schemes.

- **Platform Oversight:** Provides aggregated telemetry on registered farmers, total recommendations, and system health.
- **Advisory Management:** Enables the publication of official government schemes (e.g., PM-KISAN, Soil Health Card) and manual override of baseline APMC prices.

4. Database Architecture & Caching Strategy

Smart Krishi utilizes an enterprise persistence and caching architecture to ensure strict data integrity and high availability in rural network environments.

Feature Category	MySQL 8.0+ (Cloud Production Core)	Multi-Tier Caffeine Cache
Data Types	User Accounts, Farmer Profiles, Crop Recommendations, Historical Mandi Prices, Advisories.	Live WeatherAPI Telemetry, AGMARKNET Market Rates, Geodesic Coordinates.
Schema Rules	Strict schemas, ACID compliance, enforced referential integrity (foreign keys, cascading).	10-minute fresh TTL, 6-hour bounded stale TTL fallback.
Connection	Spring Data JPA, Hibernate.	In-Memory thread-safe caching.
Primary Key Strategy	`BIGINT AUTO_INCREMENT` for primary entities.	Normalized coordinate keys for geospatial matching.

5. Non-Functional Requirements

- **Performance & Latency:** Cached API queries (weather, crop recommendation, mandi prices) must resolve in under 100ms. Live external telemetry calls must resolve in under 500ms.
- **Security:** All passwords must be hashed using BCrypt (salt factor 10). The platform must externalize JWT secrets via environment variables, enforce strict CORS policies, and prevent SQL injection via Spring Data JPA parameterized queries.
- **Fault Tolerance:** The system shall implement circuit breakers (2 consecutive timeouts trigger a 20-second bounded backoff) when communicating with external endpoints. A 3-tier fallback architecture (Live API → 15-min Cache → Verified Admin Database) ensures uninterrupted service for market prices if AGMARKNET experiences downtime.
- **Integrity:** Weather and mandi modules strictly enforce zero simulated or fabricated values. If external APIs are unreachable and no cache exists, the system gracefully handles the failure rather than presenting false information to the farmer.